Minimum Description Length Estimation for Stochastic Processes: A Unified Framework for Statistical Learning and Data Compression

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Abstract

We develop a unified framework for consistent estimation across a broad class of stochastic processes using the minimum description length (MDL) principle. Our primary contribution is methodological: we demonstrate that a single estimation procedure achieves consistent inference for fundamentally different stochastic structures, including density estimation, Poisson processes, point processes with multiplicative intensities, continuous-time Markov chains, and diffusion processes. For each process class, we derive explicit relationships between the Hellinger distance of probability measures and the corresponding functional parameters, revealing a common mathematical structure that enables unified analysis. The MDL estimator satisfies an oracle inequality of the form $H^2(P, \hat{P}_n) \leq \inf_{Q \in \Gamma_n}(H^2(P, Q) + C_n(Q)/n) + O((\log n)/n)$, automatically balancing approximation accuracy and model complexity. We develop practical applications to universal data compression and automatic model selection, providing explicit algorithms that handle heterogeneous data sources without prior knowledge of the underlying stochastic structure.

Keywords: Minimum description length; Nonparametric estimation; Hellinger distance; Counting processes; Diffusion processes; Point processes; Model selection; Data compression; Universal coding

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Highlights:

- Unified MDL estimator applicable across five classes of stochastic processes

- Explicit Hellinger distance formulas linking measure-level and parameter-level distances
- Oracle inequality demonstrating automatic bias-complexity tradeoff
- Practical algorithms for universal compression and automatic model selection
- Theoretical foundation for handling heterogeneous data with a single procedure

1 Introduction

1.1 Motivation and Objectives

Modern data sources exhibit remarkable heterogeneity. A single application may need to process continuous signals, event sequences, state transitions, and diffusion trajectories—each traditionally requiring distinct statistical methodologies. Video compression, for instance, must handle both the temporal dynamics of scene changes (naturally modeled as point processes) and the spatial correlations within frames (often modeled through Markovian or diffusion-type structures). Similarly, in machine learning pipelines, practitioners frequently encounter mixed data types that do not fit neatly into a single probabilistic framework.

This fragmentation poses practical challenges. Developing separate estimation procedures for each data type leads to:

- duplicated algorithmic efforts across application domains,
- difficulty in building unified software systems that handle heterogeneous data streams,
- lack of principled methods for model selection when the underlying stochastic structure is unknown or evolves over time.

The present work addresses these challenges by developing a *unified estimation framework* based on the Minimum Description Length (MDL) principle. Our primary objective is not to establish new convergence rates—which are well-understood for individual model classes—but rather to demonstrate that a *single estimation principle* achieves consistent inference across fundamentally different stochastic structures, with explicit and comparable performance guarantees.

1.2 The Unification Problem

Consider the following estimation problems, each central to different application domains:

(i) **Density estimation:** Given i.i.d. samples $X_1, \ldots, X_n$ from an unknown density p, estimate p.

(ii) **Poisson process estimation:** Given event times $(T_1, \ldots, T_N)$ from a Poisson process with unknown mean measure μ, estimate μ.

(iii) **Point process intensity estimation:** Given observations of a counting process $N(t)$ with multiplicative intensity $\lambda(t) = \alpha(t)Y(t)$, estimate the baseline intensity α.

(iv) **Markov transition estimation:** Given sample paths of a continuous-time Markov chain, estimate the transition intensities $\alpha^{hj}(t)$.

(v) **Diffusion drift estimation:** Given a trajectory of the process $dX_t = \theta(X_t)dt + \sigma dW_t$, estimate the drift function θ.

Each problem has been extensively studied using tailored techniques: kernel methods and sieves for densities (Devroye and Lugosi, 2001), martingale estimating equations for point processes (Andersen et al., 1993), maximum likelihood and Girsanov transformations for diffusions (Kutoyants, 2004). However, these approaches share a common mathematical structure that has not been fully exploited: in each case, estimation reduces to optimizing a likelihood-type criterion over a space of candidate models, and consistency is governed by the metric geometry of this space.

The key insight underlying our unified framework is that the Hellinger distance provides a *universal metric* for measuring estimation accuracy across all these settings. While the functional parameters differ (densities, intensities, drift functions), the probability measures they induce can all be compared through Hellinger distance, and this comparison admits explicit formulas linking the measure-level distance to the parameter-level distance.

1.3 Contributions

This paper makes three main contributions:

1. A unified MDL estimation principle. We formulate a single minimum complexity estimator that applies uniformly to densities, Poisson processes, general point processes, Markov chains, and diffusions. The estimator takes the form

$$\hat{P}_n = \arg\min_{Q \in \Gamma_n} \left(-\log \frac{dQ}{dP_0}(\text{data}) + C_n(Q) \right), \tag{1}$$

where Γ_n is a countable model class, P_0 is a reference measure, and $C_n(Q)$ is a complexity penalty satisfying the Kraft inequality. The same algorithmic template—enumerate models, compute penalized log-likelihoods, select the minimizer—applies across all process types.

2. Explicit Hellinger distance relationships. For each process class, we derive the precise relationship between the Hellinger distance of probability measures and the natural distance on functional parameters. These relationships take the form:

$$H^2(P_\theta, P_\gamma) = 1 - \mathbb{E}_\zeta \left[\exp\left(-c \int d^2(\theta, \gamma)\, d\nu \right) \right], \tag{2}$$

where ζ is an intermediate parameter, c is a model-specific constant, $d(\theta, \gamma)$ is the L^2-type distance between parameters, and ν is an appropriate measure. Table 1 summarizes these relationships. Crucially, all formulas share similar analytic properties, enabling a unified convergence analysis.

3. Unified oracle inequality. We establish that the MDL estimator satisfies, across all process classes,

$$H^2(P, \hat{P}_n) \leq \inf_{Q \in \Gamma_n} \left(H^2(P, Q) + \frac{C_n(Q)}{n} \right) + O\left(\frac{\log n}{n} \right), \tag{3}$$

where the approximation term $H^2(P, Q)$ measures model misspecification and the complexity term $C_n(Q)/n$ penalizes model richness. This oracle inequality demonstrates that MDL estimation automatically balances the bias-variance tradeoff, achieving near-optimal performance without requiring prior knowledge of the true model's complexity.

1.4 Significance for Applications

The unified framework has direct implications for two application domains:

Universal data compression. The MDL principle originates from the observation that optimal statistical inference and optimal data compression are fundamentally equivalent (Rissanen, 1978; Cover and Thomas, 2006). An estimator achieving the bound (3) automatically yields a two-stage code with redundancy

$$R_n = \frac{1}{n} \Big(-\log \hat{P}_n(\text{data}) + \log P(\text{data}) \Big) \leq \inf_{Q \in \Gamma_n} \left(D(P \| Q) + \frac{C_n(Q)}{n} \right),$$

where $D(P \| Q)$ is the Kullback-Leibler divergence. Our results imply that a *single compression algorithm* based on MDL can achieve near-optimal compression rates for data generated by any of the five process classes considered, without prior knowledge of which class generated the data.

Automatic model selection. In many applications, the practitioner does not know *a priori* whether data should be modeled as i.i.d., as a point process, or as a diffusion. The MDL framework provides a principled solution: include models from multiple classes in Γ_n, assign appropriate complexity penalties, and let the data select the best representation. The oracle inequality (3) guarantees that this procedure performs nearly as well as if the correct model class had been known in advance.

1.5 Relation to Prior Work

The theoretical foundations of MDL estimation were laid by Rissanen (1978, 1983, 1984) and developed into a comprehensive framework by Barron and Cover (1991), Barron et al. (1998), and Grünwald (2007). Oracle inequalities of the type (3) for density estimation appear in Barron et al. (1999) and Yang and Barron (1999), with connections to model selection explored by Birgé and Massart (1998); Birgé (2006) and Hansen and Yu (2001).

For specific process classes, sieve estimation and penalized likelihood methods have been developed by Karr (1987) and McKeague (1986) for point processes, by Andersen et al. (1993) for counting processes, and by Kutoyants (2004) and Prakasa Rao (1999) for diffusions. The

Table 1: Summary of Hellinger distance relationships across process classes. In each case, $H^2(P_\theta, P_\gamma)$ denotes the squared Hellinger distance between probability measures, and the right column shows the explicit formula in terms of functional parameters.

Process Class	**Parameter**	**Hellinger Relationship**
I.i.d. density	density p	$H^2(P^n, Q^n) = 1 - (1 - H^2(P, Q))^n$
Poisson process	intensity α	$H^2(P_\alpha, P_\beta) = 1 - e^{-H^2(\alpha,\beta)}$
Point process	intensity α	$H^2(P_\alpha, P_\beta) = 1 - \mathbb{E}\left[e^{-\frac{1}{2}\int(\sqrt{\alpha}-\sqrt{\beta})^2 Y\, ds}\right]$
Markov chain	transitions α^{hj}	$H^2(P_\alpha, P_\gamma) = 1 - \mathbb{E}\left[e^{-\frac{1}{2}\sum_{h,j}\int(\sqrt{\alpha^{hj}}-\sqrt{\gamma^{hj}})^2 Y_h\, ds}\right]$
Diffusion	drift θ	$H^2(P_\theta, P_\gamma) = 1 - \mathbb{E}\left[e^{-\frac{1}{8\sigma^2}\int(\theta-\gamma)^2 dt}\right]$

role of Hellinger distance in nonparametric estimation was clarified by Le Cam (1973, 1986) and Birgé (1983).

Our contribution relative to this literature is not to improve convergence rates for any individual model class—the rates we obtain are comparable to known results—but rather to provide:

(a) a *single, explicit estimator* that works across all classes without modification,

(b) *explicit formulas* linking Hellinger distances at the measure level to parameter distances, enabling direct comparison of estimation difficulty across model types,

(c) a *unified proof technique* that highlights the common mathematical structure underlying consistency in heterogeneous settings.

To our knowledge, no prior work has systematically developed MDL estimation across densities, Poisson processes, general point processes, Markov chains, and diffusions within a single framework with explicit, comparable bounds.

1.6 Paper Organization

Section 2 introduces the statistical setting, defines the Hellinger distance and affinity, and establishes their key properties for product measures. Section 3 presents the main convergence theorem for MDL estimators in abstract form. Section 4 instantiates this theorem for each of the five process classes, deriving the explicit Hellinger relationships and stating the corresponding convergence results, with applications to compression and model selection. Section 5 contains all proofs. Section 6 discusses limitations, extensions, and open problems.

2 The Statistical Setting

2.1 Hellinger Distance and Affinity

For two probability measures P and Q on a measurable space $(\mathcal{X}, \mathcal{A})$, both dominated by a σ-finite measure μ, the *Hellinger distance* $H(P,Q)$ is defined as (Le Cam, 1986):

$$H^2(P,Q) = \frac{1}{2}\int_{\mathcal{X}} \left(\sqrt{\frac{dP}{d\mu}} - \sqrt{\frac{dQ}{d\mu}}\right)^2 d\mu. \tag{4}$$

The *Hellinger affinity* between P and Q is defined as (Kakutani, 1948):

$$\rho(P,Q) = \int_{\mathcal{X}} \sqrt{\frac{dP}{d\mu}\frac{dQ}{d\mu}}\, d\mu. \tag{5}$$

These quantities are related by:

$$H^2(P,Q) = 1 - \rho(P,Q). \tag{6}$$

Remark 2.1 (Convention)**.** *Throughout this paper, we use the convention that $H^2(P,Q) \in [0,1]$ with $H^2(P,Q) = 0$ if and only if $P = Q$, and $H^2(P,Q) = 1$ if and only if P and Q are mutually singular. This corresponds to the "squared Hellinger distance" in some references.*

2.2 Properties of Hellinger Distance

The Hellinger distance enjoys several important properties:

Lemma 2.2 (Basic Properties)**.** *The Hellinger distance satisfies:*

(i) ***Metric:*** *H is a metric on the space of probability measures.*

(ii) ***Bounds:*** $0 \leq H^2(P,Q) \leq 1$.

(iii) ***Relation to total variation:*** $H^2(P,Q) \leq \|P - Q\|_{TV} \leq \sqrt{2}H(P,Q)$.

(iv) ***Relation to KL divergence:*** $2H^2(P,Q) \leq D(P\|Q)$.

2.3 Hellinger Distance for Product Measures

A crucial property for our analysis is the behavior of Hellinger distance under products.

Lemma 2.3 (Product Measures)**.** *Let $P^n = P \otimes \cdots \otimes P$ and $Q^n = Q \otimes \cdots \otimes Q$ be n-fold product measures. Then:*

$$\rho(P^n, Q^n) = [\rho(P,Q)]^n = [1 - H^2(P,Q)]^n, \tag{7}$$

and consequently:

$$H^2(P^n, Q^n) = 1 - [1 - H^2(P,Q)]^n. \tag{8}$$

Proof. The affinity factorizes:

$$\rho(P^n, Q^n) = \int \prod_{i=1}^{n} \sqrt{\frac{dP}{d\mu}(x_i)\frac{dQ}{d\mu}(x_i)}\, d\mu^n$$
$$= \prod_{i=1}^{n} \int \sqrt{\frac{dP}{d\mu}\frac{dQ}{d\mu}}\, d\mu = [\rho(P,Q)]^n.$$

The result follows from $H^2 = 1 - \rho$. $\square$

Corollary 2.4 (Approximation for Small Distances)**.** *For $H^2(P,Q) \ll 1$:*

$$H^2(P^n, Q^n) \approx n \cdot H^2(P,Q). \tag{9}$$

This corollary explains why the complexity term in the oracle inequality (3) is scaled by $1/n$: the information from n observations accumulates linearly, while the complexity cost is fixed.

2.4 The MDL Estimation Principle

Let Γ_n be a countable collection of candidate probability measures, and let $C_n : \Gamma_n \to [0, \infty)$ be a *complexity function* satisfying the Kraft inequality:

$$\sum_{Q \in \Gamma_n} 2^{-C_n(Q)} \leq 1. \tag{10}$$

Remark 2.5 (Logarithmic base)**.** *We adopt the convention that* log *denotes the natural logarithm throughout the analysis, so that complexities $C_n(Q)$ are measured in nats and the Kraft inequality* (10) *reads $\sum_Q e^{-C_n(Q)} \leq 1$ in these units. We retain the base-*2 *form in* (10) *(and the bits unit in Section 4.6) for its coding interpretation: a change of base merely rescales $C_n(Q)$ by the constant factor* $\log 2$ *and is absorbed into the constants of the oracle inequality. All exponential and Markov inequalities below are stated with the natural base accordingly.*

The *minimum description length (MDL) estimator* is defined as:

$$\hat{P}_n = \arg\min_{Q \in \Gamma_n} \left(-\log \frac{dQ}{dP_0}(X^n) + C_n(Q) \right), \tag{11}$$

where P_0 is a fixed reference measure dominating all $Q \in \Gamma_n$, and X^n denotes the observed data.

The interpretation is:

- $-\log \frac{dQ}{dP_0}(X^n)$: code length for the data given model Q
- $C_n(Q)$: code length for describing the model Q
- The MDL estimator minimizes total description length

3 Convergence of the MDL Estimator

3.1 Main Result

Theorem 3.1 (Main Convergence Theorem). *Let $(\Omega^n, \mathcal{A}^n, P)$ be a probability space and Γ_n a countable collection of probability measures with complexity function $C_n : \Gamma_n \to [0, \infty)$ satisfying the Kraft inequality* (10)*. Let P_0 be a reference measure dominating all $Q \in \Gamma_n$.*

The MDL estimator (11) *satisfies, for any P:*

$$H^2(P, \hat{P}_n) \leq \inf_{Q \in \Gamma_n} \left(H^2(P, Q) + \frac{C_n(Q)}{n} \right) + O\left(\frac{\log n}{n} \right) \tag{12}$$

P-almost surely as $n \to \infty$.

More precisely, for any sequence $P_n \in \Gamma_n$:

$$P\left(H^2(P, \hat{P}_n) > r_n \right) \leq 2 \exp\left(-\frac{n}{2} \left(r_n - H^2(P, P_n) - \frac{C_n(P_n)}{n} \right)_+ \right) + o(1), \tag{13}$$

where $(x)_+ = \max(x, 0)$.

The proof is given in Section 5.

Remark 3.2 (On the complexity condition). *The proof of Theorem 3.1 controls the union bound through the quantity $\sum_{Q \in \Gamma_n} 2^{-C_n(Q)/2}$ (see Step 2 in Section 5), and the argument as written requires the* summability at half exponent

$$\sum_{Q \in \Gamma_n} 2^{-C_n(Q)/2} \leq 1, \tag{14}$$

which is strictly stronger than the Kraft inequality (10)*: by Cauchy–Schwarz $\sum 2^{-C_n(Q)} \leq \left(\sum 2^{-C_n(Q)/2} \right)^2$, so* (14) *implies* (10) *but not conversely. Condition* (14) *is always achievable at the cost of a factor* 2 *in the penalty: if $\tilde{C}_n$ satisfies Kraft* (10)*, then $C_n := 2\tilde{C}_n$ satisfies* (14)*, and the oracle inequality* (12) *holds with $C_n(Q)/n = 2\tilde{C}_n(Q)/n$. We state the theorem under* (14)*; the complexity functions used in Section 4 are defined with this factor absorbed.*

3.2 Interpretation

The oracle inequality (12) decomposes the estimation error into two terms:

1. **Approximation error** $H^2(P, Q)$: How well the model class Γ_n can approximate the true distribution P. This decreases as Γ_n becomes richer.

2. **Complexity penalty** $C_n(Q)/n$: The cost of model complexity, scaled by sample size. This increases with model richness.

The MDL estimator automatically balances these competing terms, achieving a tradeoff that is optimal up to logarithmic factors.

Remark 3.3 (On the $1/n$ scaling). *The factor $1/n$ in front of $C_n(Q)$ arises naturally: the complexity $C_n(Q)$ represents the total code length for the model, while n observations provide n units of information. The ratio measures the per-observation cost of model complexity.*

4 Applications to Specific Process Classes

We now instantiate the general MDL framework for each of the five process classes introduced in Section 1.

4.1 Density Estimation

Let $X_1, \ldots, X_n$ be i.i.d. observations from an unknown density p with respect to a dominating measure μ on $(E, \mathcal{E})$.

Reference measure. $P_0 = \mu^n$ (product of dominating measures).

Likelihood. For candidate density $q \in \Gamma_n$:

$$\frac{dQ^n}{d\mu^n}(X_1, \ldots, X_n) = \prod_{i=1}^{n} q(X_i). \tag{15}$$

Model class. Histogram densities on partitions:

$$\Gamma_n^{\text{hist}} = \bigcup_{m=1}^{n} \left\{ q : q = \sum_{j=1}^{m} \frac{\theta_j}{\mu(A_j)} \mathbf{1}_{A_j},\ \theta_j \in \mathcal{Q}_m,\ \sum_j \theta_j = 1 \right\}, \tag{16}$$

where $\{A_1, \ldots, A_m\}$ is a partition and $\mathcal{Q}_m$ is a finite grid of probabilities.

Complexity.

$$C_n(q) = \log^* m + (m-1)\log m, \tag{17}$$

where $\log^* m = \log m + \log\log m + \cdots$ (sum of positive terms).

MDL estimator.

$$\hat{p}_n = \arg\min_{q \in \Gamma_n^{\text{hist}}} \left(-\sum_{i=1}^{n} \log q(X_i) + C_n(q) \right). \tag{18}$$

Hellinger relationship. By Lemma 2.3:

$$H^2(P^n, Q^n) = 1 - (1 - H^2(P, Q))^n, \tag{19}$$

where $H^2(P,Q) = \frac{1}{2}\int(\sqrt{p} - \sqrt{q})^2 d\mu$.

Theorem 4.1 (Density Estimation Convergence)**.** *The MDL estimator* (18) *satisfies:*

$$H^2(p, \hat{p}_n) \le \inf_{m \ge 1} \left(\inf_{q \in \Gamma_m} H^2(p,q) + \frac{\log^* m + (m-1)\log m}{n} \right) + O\left(\frac{\log n}{n}\right) \tag{20}$$

P-almost surely.

4.2 Poisson Process Estimation

Let $(T_1, \ldots, T_N)$ be event times from a Poisson process on $E = [0, 1]$ with unknown mean measure μ having intensity $\alpha = d\mu/d\lambda$ with respect to Lebesgue measure λ.

Reference measure. P_0: homogeneous Poisson process with unit intensity.

Likelihood. For candidate intensity β:

$$\frac{dP_\beta}{dP_0}(T_1, \ldots, T_N) = \exp\left(\int_0^1 (1 - \beta(s))ds\right) \prod_{i=1}^{N} \beta(T_i). \tag{21}$$

Model class. Piecewise constant intensities:

$$\Gamma_n^{\text{Poisson}} = \bigcup_{m=1}^{n} \left\{\beta : \beta(t) = \sum_{j=1}^{m} \beta_j \mathbf{1}_{[(j-1)/m, j/m)}(t),\ \beta_j \in \mathcal{G}_m\right\}, \tag{22}$$

where $\mathcal{G}_m = \{k/m : k = 1, \ldots, mn\}$.

Complexity.

$$C_n(\beta) = \log^* m + m \log(mn). \tag{23}$$

MDL estimator.

$$\hat{\alpha}_n = \arg \min_{\beta \in \Gamma_n^{\text{Poisson}}} \left(\int_0^1 \beta(s)ds - \sum_{i=1}^{N} \log \beta(T_i) + C_n(\beta)\right). \tag{24}$$

Hellinger relationship.

Lemma 4.2 (Hellinger Distance for Poisson Processes)**.** *Let P_α and P_β be distributions of Poisson processes with intensities α and β. Then:*

$$H^2(P_\alpha, P_\beta) = 1 - \exp\left(-H^2(\alpha, \beta)\right), \tag{25}$$

where $H^2(\alpha, \beta) = \frac{1}{2}\int_0^1 (\sqrt{\alpha(s)} - \sqrt{\beta(s)})^2 ds$.

Theorem 4.3 (Poisson Process Convergence)**.** *The MDL estimator* (24) *satisfies:*

$$H^2(\alpha, \hat{\alpha}_n) \leq \inf_{\beta \in \Gamma_n^{Poisson}} \left(H^2(\alpha, \beta) + \frac{C_n(\beta)}{n}\right) + O\left(\frac{\log n}{n}\right) \tag{26}$$

P-almost surely, where $n = \mathbb{E}[N] = \int_0^1 \alpha(t)dt$.

4.3 Point Process with Multiplicative Intensity

Let $N(t)$, $t \in [0, T]$, be a counting process with intensity $\lambda(t) = \alpha(t)Y(t)$, where $Y(t)$ is an observable, predictable process and α is the unknown baseline intensity (Andersen et al., 1993).

Reference measure. P_0: the distribution when $\alpha \equiv 1$.

Likelihood. For candidate baseline intensity β:

$$\frac{dP_\beta}{dP_0} = \exp\left(\int_0^T (1-\beta(s))Y(s)ds + \int_0^T \log\beta(s)\, dN(s)\right). \tag{27}$$

Model class. Piecewise constant baseline intensities on dyadic partitions:

$$\Gamma_n^{\mathrm{PP}} = \bigcup_{k=0}^{\lfloor \log_2 n \rfloor} \left\{\beta : \beta(t) = \sum_{j=1}^{2^k} \beta_j \mathbf{1}_{I_{kj}}(t),\ \beta_j \in \mathcal{G}_{2^k}\right\}, \tag{28}$$

where $I_{kj} = [(j-1)T/2^k, jT/2^k)$.

Complexity.

$$C_n(\beta) = k + 2^k \log n \quad \text{for } \beta \text{ with } 2^k \text{ pieces.} \tag{29}$$

MDL estimator.

$$\hat{\alpha}_n = \arg\min_{\beta \in \Gamma_n^{\mathrm{PP}}} \left(\int_0^T \beta(s)Y(s)ds - \int_0^T \log\beta(s)\, dN(s) + C_n(\beta)\right). \tag{30}$$

Hellinger relationship.

Lemma 4.4 (Hellinger Distance for Point Processes)**.** *Let P_α and P_β be point process distributions with baseline intensities α and β. Then:*

$$H^2(P_\alpha, P_\beta) = 1 - \mathbb{E}_{P_{\sqrt{\alpha\beta}}}\left[\exp\left(-\frac{1}{2}\int_0^T (\sqrt{\alpha(s)} - \sqrt{\beta(s)})^2 Y(s)ds\right)\right]. \tag{31}$$

Theorem 4.5 (Point Process Convergence)**.** *Under the condition* $\inf_t \mathbb{E}[Y(t)] \geq c > 0$*, the MDL estimator* (30) *satisfies:*

$$\int_0^T (\sqrt{\alpha(t)} - \sqrt{\hat{\alpha}_n(t)})^2 dt \lesssim \inf_{\beta \in \Gamma_n^{PP}} \left(\int_0^T (\sqrt{\alpha} - \sqrt{\beta})^2 dt + \frac{C_n(\beta)}{n}\right) \tag{32}$$

P-almost surely.

4.4 Markov Transition Model

Consider n independent realizations of a continuous-time Markov chain on state space $\{1, \ldots, M\}$ with transition intensities $\alpha^{hj}(t)$ for $h \neq j$ (Aalen, 1978).

Reference measure. P_0: unit transition intensities.

Likelihood. For candidate intensities $\gamma = (\gamma^{hj})$:

$$\frac{dP_\gamma}{dP_0} = \prod_{h \neq j} \exp\left(\int_0^T (1 - \gamma^{hj}(s)) Y_h^{(n)}(s) ds + \int_0^T \log \gamma^{hj}(s)\, dN_{hj}^{(n)}(s) \right), \tag{33}$$

where $Y_h^{(n)} = \sum_{i=1}^n Y_h^{(i)}$ and $N_{hj}^{(n)} = \sum_{i=1}^n N_{hj}^{(i)}$.

Model class. Piecewise constant transition intensities:

$$\Gamma_n^{\text{Markov}} = \left\{ \gamma : \gamma^{hj}(t) = \sum_{\ell=1}^m \gamma_\ell^{hj} \mathbf{1}_{I_\ell}(t),\ \gamma_\ell^{hj} \in \mathcal{G}_m \right\}. \tag{34}$$

Complexity.

$$C_n(\gamma) = \log^* m + M(M-1) \cdot m \cdot \log n. \tag{35}$$

MDL estimator.

$$\hat{\alpha}_n = \arg \min_{\gamma \in \Gamma_n^{\text{Markov}}} \left(\sum_{h \neq j} \left[\int_0^T \gamma^{hj}(s) Y_h^{(n)}(s) ds - \int_0^T \log \gamma^{hj}(s)\, dN_{hj}^{(n)}(s) \right] + C_n(\gamma) \right). \tag{36}$$

Hellinger relationship.

Lemma 4.6 (Hellinger Distance for Markov Chains)**.**

$$H^2(P_\alpha, P_\gamma) = 1 - \mathbb{E}_{P_{\sqrt{\alpha\gamma}}} \left[\exp\left(-\frac{1}{2} \sum_{h \neq j} \int_0^T (\sqrt{\alpha^{hj}(s)} - \sqrt{\gamma^{hj}(s)})^2 Y_h(s) ds \right) \right]. \tag{37}$$

4.5 Diffusion Process Drift Estimation

Let X_t satisfy $dX_t = \theta(X_t)dt + \sigma dW_t$ on $[0, T]$, with $\sigma > 0$ known (Kutoyants, 2004).

Reference measure. P_0: Wiener measure ($\theta \equiv 0$).

Likelihood. By Girsanov's theorem, for candidate drift γ:

$$\frac{dP_\gamma}{dP_0} = \exp\left(\frac{1}{\sigma^2} \int_0^T \gamma(X_t) dX_t - \frac{1}{2\sigma^2} \int_0^T \gamma^2(X_t) dt \right). \tag{38}$$

Model class. Piecewise linear drift functions:

$$\Gamma_n^{\text{Diff}} = \bigcup_{m=1}^n \left\{ \gamma : \gamma(x) = \sum_{j=1}^m (a_j + b_j x) \mathbf{1}_{I_j}(x),\ a_j, b_j \in \mathcal{G}_m \right\}. \tag{39}$$

Complexity.

$$C_n(\gamma) = \log^* m + 2m \log n. \tag{40}$$

MDL estimator.

$$\hat{\theta}_n = \arg\min_{\gamma\in\Gamma_n^{\mathrm{Diff}}} \left(\frac{1}{2\sigma^2} \int_0^T \gamma^2(X_t)dt - \frac{1}{\sigma^2} \int_0^T \gamma(X_t)dX_t + C_n(\gamma) \right). \tag{41}$$

Hellinger relationship.

Lemma 4.7 (Hellinger Distance for Diffusions)**.**

$$H^2(P_\theta, P_\gamma) = 1 - \mathbb{E}_{P_{\frac{\theta+\gamma}{2}}} \left[\exp\left(-\frac{1}{8\sigma^2} \int_0^T (\theta(X_t) - \gamma(X_t))^2 dt \right) \right]. \tag{42}$$

Theorem 4.8 (Diffusion Convergence)**.** *With n independent replications on $[0, T]$, the MDL estimator* (41) *satisfies:*

$$\frac{1}{T} \int_0^T (\theta(X_t) - \hat{\theta}_n(X_t))^2 dt \lesssim \inf_{\gamma\in\Gamma_n^{Diff}} \left(\frac{1}{T} \int_0^T (\theta-\gamma)^2 dt + \frac{C_n(\gamma)}{nT} \right) \tag{43}$$

P-almost surely.

4.6 Application: Universal Data Compression

The MDL framework provides a principled approach to data compression that adapts to the underlying stochastic structure.

4.6.1 Two-Stage Coding

A two-stage code for data $\mathbf{X}$ consists of:

1. **Model code:** Encode model $\hat{Q}$ using $C_n(\hat{Q})$ bits.
2. **Data code:** Encode data given model using $-\log_2 \hat{Q}(\mathbf{X})$ bits.

Total code length: $L(\mathbf{X}) = -\log_2 \hat{Q}(\mathbf{X}) + C_n(\hat{Q})$.

Theorem 4.9 (Universal Compression)**.** *The MDL compressor achieves expected code length:*

$$\mathbb{E}_P[L(\mathbf{X})] \le H(P) + \inf_{Q\in\Gamma} (D(P\|Q) + C(Q)) + O(1), \tag{44}$$

where $H(P)$ is the entropy of P.

4.6.2 Universal Compression Algorithm

Algorithm 1 presents a universal compression scheme handling heterogeneous data.

4.7 Application: Automatic Model Selection

When the data-generating mechanism is unknown, MDL provides principled model selection.

Theorem 4.10 (Model Selection Consistency)**.** *If P belongs to class $\mathcal{C}_{k^*}$ and is separated from other classes ($\inf_{Q\in\Gamma^{(k)}} H^2(P,Q) \ge \delta > 0$ for $k \ne k^*$), then Algorithm 2 satisfies $\mathbb{P}(\hat{k} = k^*) \to 1$.*

Algorithm 1 Universal MDL Compression

Require: Data sequence $\mathbf{X}$, maximum complexity K
Ensure: Compressed representation

$\Gamma \leftarrow \emptyset$
// **Add models from each process class**
for each process class $\mathcal{C} \in \{\text{Density, Poisson, PP, Markov, Diffusion}\}$ **do**
  Add models from $\mathcal{C}$ with complexity up to K to Γ
  Adjust complexity: $C(Q) \leftarrow C_{\mathcal{C}}(Q) + \log 5$
end for
// **MDL selection**
$\hat{Q} \leftarrow \arg\min_{Q \in \Gamma}(-\log Q(\mathbf{X}) + C(Q))$
// **Encode**
ModelCode $\leftarrow$ Encode($\hat{Q}$)
DataCode $\leftarrow$ ArithmeticEncode($\mathbf{X}$, $\hat{Q}$)
return (ModelCode, DataCode)

Algorithm 2 Automatic Model Class Selection

Require: Data $\mathbf{X}$, candidate classes $\mathcal{C}_1, \ldots, \mathcal{C}_K$
Ensure: Selected class $\hat{k}$, model $\hat{Q}$

for $k = 1$ to K **do**
  $\hat{Q}_k \leftarrow \arg\min_{Q \in \Gamma_n^{(k)}}(-\log Q(\mathbf{X}) + C_n^{(k)}(Q))$
  $L_k \leftarrow -\log \hat{Q}_k(\mathbf{X}) + C_n^{(k)}(\hat{Q}_k) + \log K$
end for
$\hat{k} \leftarrow \arg\min_k L_k$
return $(\hat{k}, \hat{Q}_{\hat{k}})$

5 Proofs

5.1 Notation and Conventions

Throughout this section:

- $H^2(P, Q) = 1 - \rho(P, Q)$ where $\rho(P, Q) = \int \sqrt{dP/d\mu \cdot dQ/d\mu}\, d\mu$
- $C_n : \Gamma_n \to [0, \infty)$ satisfies $\sum_{Q \in \Gamma_n} 2^{-C_n(Q)} \leq 1$
- $(x)_+ = \max(x, 0)$

5.2 Technical Lemmas

Lemma 5.1 (Likelihood Ratio Bound)**.** *For probability measures P, S with $P \ll S$, and any $a > 0$:*

$$P\left(\frac{dP}{dS} \leq a\right) \leq \min\left\{a, \sqrt{a} \cdot (1 - H^2(P, S))\right\}. \tag{45}$$

Proof. **First bound:** By Markov's inequality,

$$P\left(\frac{dS}{dP} \geq \frac{1}{a}\right) \leq a \cdot \mathbb{E}_P\left[\frac{dS}{dP}\right] = a.$$

Second bound: Applying Markov to $\sqrt{dS/dP}$:

$$P\left(\frac{dP}{dS} \leq a\right) \leq \sqrt{a} \cdot \mathbb{E}_P\left[\sqrt{\frac{dS}{dP}}\right] = \sqrt{a} \cdot \rho(P,S) = \sqrt{a} \cdot (1 - H^2(P,S)).$$

$\square$

Lemma 5.2 (Penalized Likelihood Comparison)**.** *Let $Q, S \in \Gamma_n$ and define*

$$A(Q,S) = \left\{\frac{dQ}{dS} \geq 2^{C_n(Q)-C_n(S)}\right\}.$$

Then for any $a > 0$:

$$P\left(A(Q,S) \cap \left\{\frac{dP}{dS} \leq a\right\}\right) \leq a \cdot 2^{\frac{1}{2}(C_n(S)-C_n(Q))}. \tag{46}$$

Proof. Using exponential bounds with $h = 1/2$:

$$\begin{aligned} P\left(A(Q,S) \cap \left\{\frac{dP}{dS} \leq a\right\}\right) &\leq 2^{\frac{1}{2}(C_n(S)-C_n(Q))} \int_{\{dP/dS \leq a\}} \sqrt{\frac{dQ}{dS}}\, dP \\ &\leq 2^{\frac{1}{2}(C_n(S)-C_n(Q))} \cdot a \int \sqrt{\frac{dQ}{dS}}\, dS \\ &= a \cdot 2^{\frac{1}{2}(C_n(S)-C_n(Q))}. \end{aligned}$$

$\square$

5.3 Proof of Main Theorem

Proof of Theorem 3.1. Let $r_n > 0$, $P_n \in \Gamma_n$, and $a_n = n^{-(1+\epsilon)}$ for $\epsilon > 0$.

Define:

- $U_n = \{Q \in \Gamma_n : H^2(P,Q) > r_n\}$
- $D_n = \{dP/dP_n \leq a_n\}$

Step 1: Event decomposition.

$$P(\hat{P}_n \in U_n) \leq P\left(\bigcup_{Q \in U_n} A(Q, P_n) \cap D_n\right) + P(D_n^c).$$

Step 2: First term. By Lemma 5.2 and union bound:

$$
\begin{aligned}
P\left(\bigcup_{Q\in U_n} A(Q,P_n)\cap D_n\right) &\le \sum_{Q\in U_n} a_n \cdot 2^{\frac{1}{2}(C_n(P_n)-C_n(Q))} \\
&\le a_n \cdot 2^{\frac{C_n(P_n)}{2}} \sum_{Q\in\Gamma_n} 2^{-\frac{C_n(Q)}{2}} \\
&\le a_n \cdot 2^{\frac{C_n(P_n)}{2}}.
\end{aligned}
$$

Step 3: Second term. This step requires structural assumptions on (P, P_n). Suppose $P = p^{\otimes n}$ and $P_n = p_n^{\otimes n}$ are n-fold product measures (the i.i.d. / replicated-sample setting of Section 4), so that the affinity factorises, $\rho(P, P_n) = (1-H^2(p, p_n))^n$ by Lemma 2.3. Applying the second bound of Lemma 5.1 with $S = P_n$ and $a = a_n$ gives

$$
P(D_n) = P\left(\frac{dP}{dP_n} \le a_n\right) \le \sqrt{a_n}\,\rho(P,P_n) = \sqrt{a_n}\,(1 - H^2(p,p_n))^n,
$$

and the complementary "typical-set" bound used below,

$$
P(D_n^c) \le \frac{(1-H^2(p,p_n))^n}{\sqrt{a_n}},
$$

follows by the same Cramér–Chernoff control of the log-likelihood ratio $\log(dP_n/dP)$, whose cumulant generating function is governed by the per-coordinate affinity. This is the Barron–Cover typical-set argument (Barron and Cover, 1991); it uses the product (more generally, factorising-affinity) structure in an essential way.

Remark 5.3 (Scope of Step 3)**.** *The factorisation of the affinity, and hence the exponential bound on $P(D_n^c)$, holds verbatim only when the data decomposes into n independent (or independent replicated) components, as in density estimation, the n-replication Markov and diffusion models, and the high-intensity Poisson regime. For the genuinely continuous-time single-path models (point process and single-trajectory diffusion) the same conclusion requires a martingale exponential-inequality substitute for Lemma 2.3; we use n to denote the relevant information amount ($\mathbb{E}[N]$ or the number of replications) and treat those cases under the corresponding independence hypotheses stated in Section 4. The argument should therefore be read as a Barron–Cover-type sketch under factorising affinity, not as a single proof covering all five classes uniformly without further hypotheses.*

Step 4: Combining.

$$
P(\hat{P}_n \in U_n) \le n^{-(1+\epsilon)} \cdot 2^{\frac{C_n(P_n)}{2}} + n^{\frac{1+\epsilon}{2}} \cdot e^{-nH^2(P,P_n)}.
$$

Step 5: Choice of r_n. Set $r_n = H^2(P,P_n) + \frac{C_n(P_n)}{n} + \frac{(2+\epsilon)\log n}{n}$.
Both terms are summable, so by Borel-Cantelli:

$$
H^2(P,\hat{P}_n) \le H^2(P,P_n) + \frac{C_n(P_n)}{n} + O\left(\frac{\log n}{n}\right) \quad P\text{-a.s.}
$$

Optimizing over $P_n \in \Gamma_n$ yields the result. □

5.4 Proofs of Hellinger Relationships

Proof of Lemma 4.2. Let P_0 be a Poisson process with unit intensity. The Hellinger affinity is:

$$\begin{aligned}\rho(P_\alpha, P_\beta) &= \mathbb{E}_{P_0}\left[\sqrt{\frac{dP_\alpha}{dP_0}\frac{dP_\beta}{dP_0}}\right] \\ &= \mathbb{E}_{P_0}\left[e^{1-\frac{1}{2}(\mu_\alpha+\mu_\beta)}\prod_{i=1}^{N}\sqrt{\alpha(T_i)\beta(T_i)}\right],\end{aligned}$$

where $\mu_\alpha = \int \alpha\, ds$.

Using the Laplace functional of Poisson processes:

$$\begin{aligned}\rho(P_\alpha, P_\beta) &= \exp\left(\int_0^1 \sqrt{\alpha(s)\beta(s)}\, ds - \frac{1}{2}\int_0^1 (\alpha(s)+\beta(s))ds\right) \\ &= \exp\left(-\frac{1}{2}\int_0^1 (\sqrt{\alpha(s)} - \sqrt{\beta(s)})^2 ds\right) \\ &= \exp(-H^2(\alpha, \beta)).\end{aligned}$$

Thus $H^2(P_\alpha, P_\beta) = 1 - e^{-H^2(\alpha,\beta)}$. □

Proof of Lemma 4.4. The likelihood is $L_\alpha = \exp(\int(1-\alpha)Y\, ds + \int \log\alpha\, dN)$.

Computing $\sqrt{L_\alpha L_\beta}$:

$$\begin{aligned}\sqrt{L_\alpha L_\beta} = \exp&\left(\int(1-\zeta)Y\, ds + \int \log\zeta\, dN\right) \\ &\times \exp\left(-\frac{1}{2}\int(\sqrt{\alpha}-\sqrt{\beta})^2 Y\, ds\right),\end{aligned}$$

where $\zeta = \sqrt{\alpha\beta}$.

The first factor is L_ζ. Taking expectations under P_0:

$$\rho(P_\alpha, P_\beta) = \mathbb{E}_{P_\zeta}\left[\exp\left(-\frac{1}{2}\int(\sqrt{\alpha}-\sqrt{\beta})^2 Y\, ds\right)\right].$$

□

Proof of Lemma 4.7. By Girsanov, $L_\theta = \exp(\frac{1}{\sigma^2}\int \theta\, dX - \frac{1}{2\sigma^2}\int \theta^2\, dt)$.

Let $\zeta = (\theta+\gamma)/2$. Then:

$$\begin{aligned}\sqrt{L_\theta L_\gamma} &= L_\zeta \cdot \exp\left(\frac{1}{8\sigma^2}\int(\theta+\gamma)^2 dt - \frac{1}{4\sigma^2}\int(\theta^2+\gamma^2)dt\right) \\ &= L_\zeta \cdot \exp\left(-\frac{1}{8\sigma^2}\int(\theta-\gamma)^2 dt\right).\end{aligned}$$

Thus:

$$\rho(P_\theta, P_\gamma) = \mathbb{E}_{P_\zeta}\left[\exp\left(-\frac{1}{8\sigma^2}\int(\theta-\gamma)^2 dt\right)\right].$$

□

6 Discussion

6.1 Summary of Contributions

We have developed a unified framework for statistical estimation across multiple stochastic process classes using MDL. The main contributions are:

(i) A single MDL estimator achieving consistent estimation across densities, Poisson processes, point processes, Markov chains, and diffusions.

(ii) Explicit Hellinger distance formulas sharing a common structure:

$$H^2(P_\theta, P_\gamma) = 1 - \mathbb{E}_\zeta \left[\exp\left(-c \int d^2(\theta, \gamma)\, d\nu \right) \right].$$

(iii) A unified oracle inequality demonstrating automatic bias-complexity tradeoff.

(iv) Practical algorithms for universal compression and model selection.

6.2 Limitations

Rates of convergence. The rates obtained are not always minimax optimal. For specific model classes, specialized methods may achieve better rates. Our goal was universality, not optimality for individual classes. The gap is typically a logarithmic factor: it is an artefact of the description-length penalty (the per-coefficient quantization cost $\log n$ in, e.g., $C_n(\beta) = k + 2^k \log n$), not of the estimation problem, and it can be removed by a penalized-projection refinement that pays the model's metric complexity instead (Remark 6.2).

Computational complexity. Exact MDL optimization over large model classes may be computationally intensive. Greedy approximations (Algorithm 1) work well in practice but lack formal guarantees.

Independence assumptions. Several results assume independent observations or replications. Extensions to dependent data require additional technical machinery.

Model class specification. Results assume the true distribution can be approximated within Γ_n. Model misspecification leads to convergence to the best approximation, which may differ from the truth.

6.3 Extensions

Dependent observations. For β-mixing sequences, the effective sample size becomes $n_{\text{eff}} = n/(1 + 2\sum_k \beta_k)$, and the oracle inequality adjusts accordingly.

High-dimensional parameters. For sparse models, complexity penalties should reflect sparsity: $C_n(\theta) = \|\theta\|_0 \log(d/\|\theta\|_0) + \|\theta\|_0 \log n$.

Online estimation. Sequential MDL processes observations incrementally: $\hat{Q}_t = \arg\min_Q(-\sum_{i=1}^t \log Q($
$C(Q))$.

Bayesian approaches. Instead of model selection, Bayesian MDL computes mixture distributions with posterior weights $w(Q|X^n) \propto Q(X^n)2^{-C_n(Q)}$.

6.4 Open Problems

Open Problem 6.1 (Minimax optimality)**.** *Under what conditions does MDL achieve minimax optimal rates? Can logarithmic factors be removed?*

Remark 6.2 (Removing the logarithmic factor: a partial answer)**.** *For the density/sieve case the answer is now known. The* $\log n$ *in the piecewise complexity* $C_n(\beta) = k + 2^k \log n$ *(Eq.* (29)*) is a* coding artefact*: it is the per-piece cost of quantizing* 2^k *continuous coefficients into a prefix code. An estimator that fits those coefficients* without *quantizing them—the orthogonal* L^2*-projection (histogram / piecewise-polynomial) estimator onto the* 2^k*-piece model—pays instead the model's metric complexity* $2^k/n$*, with* no *logarithm; and the penalized version selecting* k *by* $\mathrm{pen}(k) \asymp 2^k/n$ *attains the exact minimax rate* $n^{-2s/(2s+1)}$ adaptively*, by the Birgé–Massart oracle inequality (Birgé and Massart, 1998; Barron et al., 1999). The mechanism is that the geometric growth of the dimensions* 2^k *makes the selection weights summable with exponents* $\asymp 2^k$*, so no* log *re-enters. Thus the description-length MDL is* near-minimax *(optimal up to* log*), while its projection counterpart is* exactly *minimax. What remains open is the general-process version (Poisson, Markov, diffusion): whether a single contrast-based refinement removes the* log *uniformly across the process classes treated here.*

Open Problem 6.3 (Model selection across classes)**.** *Can separation conditions in Theorem 4.10 be weakened? What happens when multiple classes provide equally good approximations?*

Open Problem 6.4 (General semimartingales)**.** *Can Hellinger formulas be unified under a general semimartingale framework, extending to Lévy processes and jump-diffusions?*

Open Problem 6.5 (Efficient computation)**.** *For which model classes does polynomial-time exact MDL optimization exist? What approximation guarantees can greedy algorithms achieve?*

6.5 Concluding Remarks

The MDL principle provides a powerful approach to statistical inference that naturally balances model fit and complexity. By developing this principle across multiple stochastic process classes, we have shown that:

1. A single estimation procedure achieves consistent inference for fundamentally different data-generating mechanisms.

2. The Hellinger distance provides a universal metric with explicit formulas linking measure-level and parameter-level distances.

3. The framework has immediate applications to universal compression and automatic model selection.

While rates may not always be optimal for specific classes, the universality of the approach is valuable for modern applications involving heterogeneous data sources.

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Declaration of Competing Interests

The author declares no competing interests.

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